

MR AHMED ABD EL-MOTAAL**Lesson 3-1: Similarity of Polygons**

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COMPLETE LESSON

Student Name: _____ **Class / Group:** _____ **Date:** ____ / ____ / 2026 **Evaluation Score:** [____ / 10]

 Prerequisites & Diagnostic Review

Review foundational concepts required before studying polygon similarity: ratios, sum of quadrilateral angles, and quadrilateral properties.

 Prerequisite Review **Ratio & Proportionality**

A ratio compares two quantities $\frac{a}{b}$. Two ratios are equal $\frac{a}{b} = \frac{c}{d} = k$ if they simplify to the same constant factor k . Cross-multiplication yields $a \times d = b \times c$.

 Quadrilateral Angle Sum

The sum of the measures of the interior angles of any convex quadrilateral is always 360° . If 3 angles are known, the 4th is: $m(\angle 4) = 360^\circ - (\angle_1 + \angle_2 + \angle_3)$.

 Special Quadrilateral Properties

In a parallelogram/rhombus: opposite angles are equal ($m(\angle A) = m(\angle C)$), and consecutive angles are supplementary ($m(\angle A) + m(\angle B) = 180^\circ$). In rectangles and squares, all 4 angles are 90° .

 Diagnostic Check: Test Your Readiness

QUESTION 1 OF 3

1. In any quadrilateral $ABCD$, if $m(\angle A) = 90^\circ$, $m(\angle B) = 120^\circ$, and $m(\angle C) = 45^\circ$, what is $m(\angle D)$?

☐ A 105° ☐ B 115° ☐ C 95° ☐ D 125°

QUESTION 2 OF 3

2. In rhombus $ABCD$, if $m(\angle A) = 110^\circ$, then the consecutive angle $m(\angle B) =$

☐ A 70° ☐ B 80° ☐ C 110° ☐ D 90°

QUESTION 3 OF 3

3. If $\frac{12}{8} = \frac{x}{6}$, then $x =$

☐ A 10☐ B 12☐ C 8☐ D 9

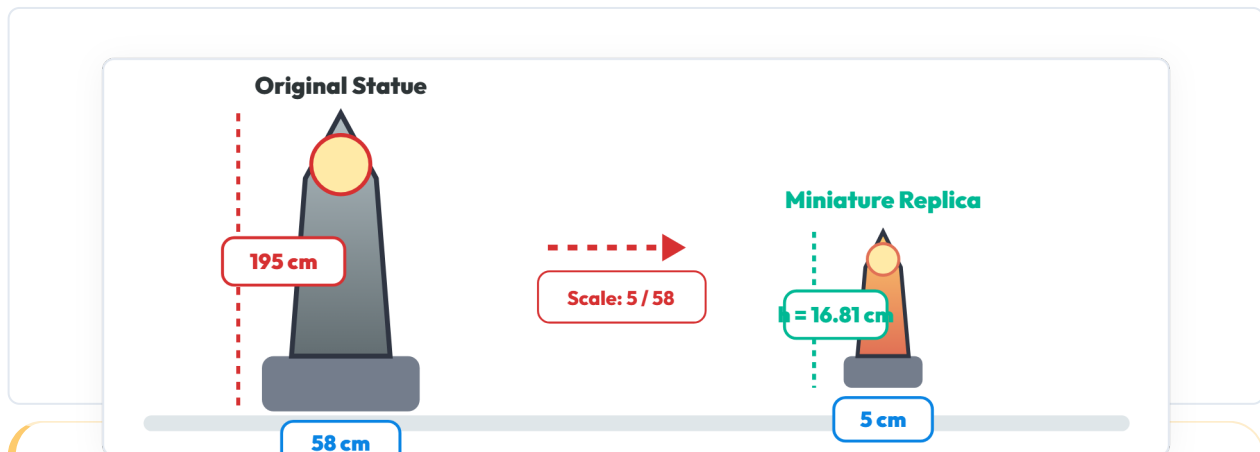
Real-World Connections (Mathematics in Everyday Life)

Precision Plots & Modeling

Egyptian Heritage & Archaeological Modeling • Statue of King Thutmose III

STEM Real-World Application

The ancient Egyptians carved monumental statues for grand temples. Modern museum curators and 3D preservation labs rely on geometric similarity to produce exact miniature replicas with authentic proportions.



Mathematical Connection & STEM Analysis:

Mathematical Proportional Modeling:

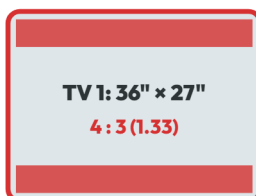
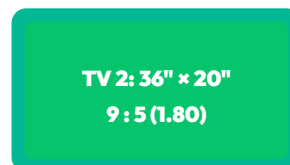
Because the miniature replica is geometrically similar to the original statue (height 195 cm, width 58 cm), corresponding linear dimensions are directly proportional:

$$\frac{\text{Height of Replica}}{\text{Height of Original}} = \frac{\text{Width of Replica}}{\text{Width of Original}} \implies \frac{h}{195} = \frac{5}{58}$$

Applying cross-multiplication yields $58h = 975 \implies h = \frac{975}{58} \approx 16.81$ cm. This uniform scale factor preserves the authentic pharaonic proportions with zero distortion.

Digital Media & Optics • Cinema Screen vs TV Aspect Ratios**STEM Real-World Application**

Modern entertainment screens rely strictly on geometric similarity (aspect ratios) to display high-definition films without image distortion or black letterbox cropping.

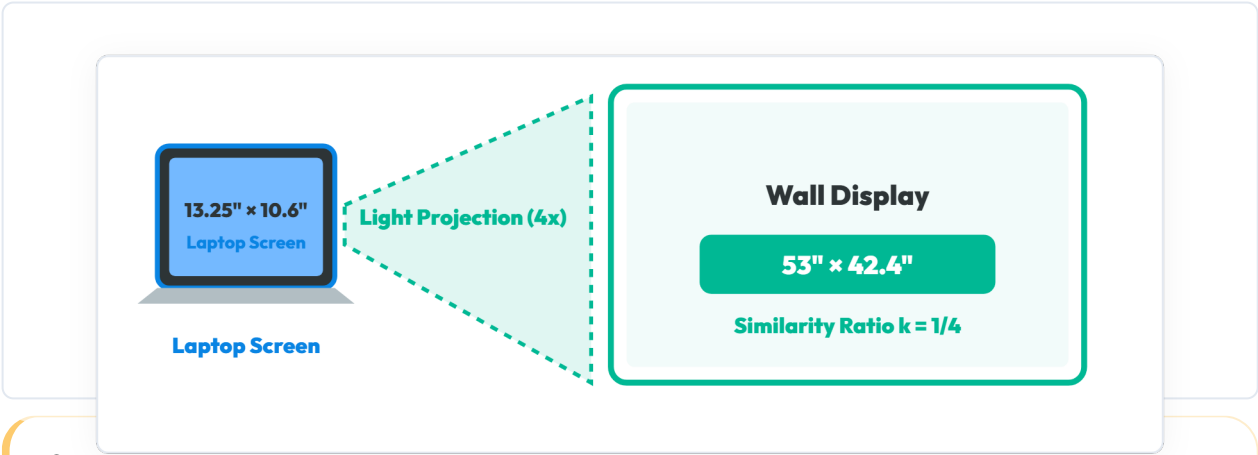
**Original Cinema Screen****Not Similar (Letterbox)****Similar (Full Display)****💡 Mathematical Connection & STEM Analysis:****Geometric Aspect Ratio Proof:**

A cinema screen measuring $45 \text{ ft} \times 25 \text{ ft}$ has an aspect ratio of $\frac{45}{25} = \frac{9}{5} = 1.80$. A TV screen measuring $36 \text{ in} \times 20 \text{ in}$ possesses the identical ratio $\frac{36}{20} = 1.80$, making it geometrically similar to the cinema screen. Consequently, TV 2 displays the cinema film with complete fidelity, while older $4 : 3$ monitors ($\frac{36}{27} \approx 1.33$) force image compression or letterbox bars.

🖥 Classroom & Office Technology • Laptop to Classroom Projector Wall

🌐 STEM Real-World Application

Digital optical projectors magnify compact computer screens onto wide presentation walls using geometric dilation.



💡 Mathematical Connection & STEM Analysis:

Step-by-Step Similarity Verification:

Both screens are rectangles, ensuring all corresponding interior angles are 90° . Comparing dimensions yields:

$$\text{Width Ratio} = \frac{13.25}{53} = \frac{1}{4} = 0.25, \quad \text{Height Ratio} = \frac{10.6}{42.4} = \frac{1}{4} = 0.25$$

Because all angles are congruent and side lengths share the uniform scale factor $k = \frac{1}{4}$, the laptop and projected wall display are geometrically similar, providing crisp $4\times$ magnification.

📌 Essential Curriculum Standards

Definitions & Standard Forms

1

Two Simultaneous Conditions

Equality of angles alone is NOT sufficient (e.g. rectangle vs square), nor is proportionality of sides alone sufficient (e.g. rhombus vs square). Both must hold true!

2

Vertex Correspondence Order

Vertices must be written in exact corresponding order: $ABCD \sim EFGH \implies A \leftrightarrow E, B \leftrightarrow F, C \leftrightarrow G, D \leftrightarrow H$. Side ratios follow immediately: $\frac{AB}{EF} = \frac{BC}{FG} = \dots$

3

Similarity Ratio / Scale Factor k

If $k > 1$, polygon 1 is an Enlargement of polygon 2. If $0 < k < 1$, it is a Reduction. If $k = 1$, the polygons are Congruent.

4

Transitivity Principle

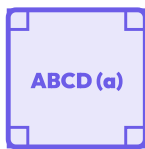
Two polygons similar to a third polygon are similar to each other. In addition, all congruent polygons are similar ($k = 1$), but similar polygons are NOT necessarily congruent.



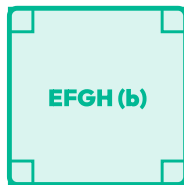
Special Quadrilateral Criteria & Regular Polygons

Geometric Analysis

■ 1. All Squares Are Similar



Square 1

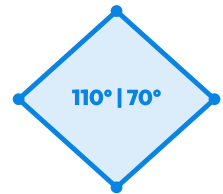
Square 2 ($k = a/b$)

- All 4 angles are right angles ($90^\circ = 90^\circ$).
- All 4 sides are equal, so side ratios $\frac{a}{b} = \frac{a}{b} = \dots = k$ are always constant:

◆ 2. Rhombuses: One Angle Suffices



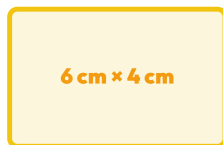
Rhombus 1



Rhombus 2 (Same Angles)

- Two rhombuses are similar if **one pair of corresponding angles is equal** ($m(\angle A) = m(\angle E)$).
- Because all 4 sides of each rhombus are equal, side proportionality is guaranteed automatically!

3. Rectangles: Proportional Dimensions



Rectangle 1

Rectangle 2 ($k = 2$)

- All interior angles are 90° automatically.

- Two rectangles are similar if and only if their **two adjacent dimensions are proportional** : $\frac{\text{Length}_1}{\text{Length}_2} = \frac{\text{Width}_1}{\text{Width}_2}$.

4. Regular Polygons: Same Side Count

Pentagon 1
5-gon ($s = 4$)Pentagon 2
5-gon ($s = 8$) ✓ Similar

- Regular polygons having the **same number of sides are always similar** (e.g. all equilateral triangles, all regular hexagons).
- **Spot the Error:** Regular polygons with different numbers of sides (e.g. regular triangle and regular pentagon) are **NEVER ! similar**

Concept Flashcard & Strategy Guide

Idea 1: Proving Similarity & Calculating the Scale Factor (k)



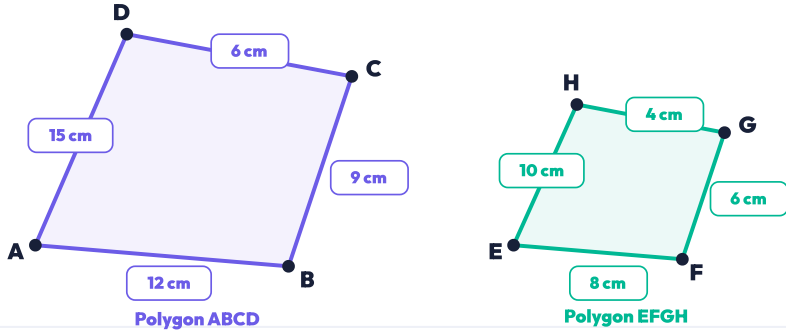
Mastery Strategy: To prove two polygons are similar, verify that every pair of corresponding angles is congruent, and calculate the ratio $\frac{\text{Side}_1}{\text{Side}_2}$ for all pairs of corresponding sides. If all ratios simplify to the exact same number k , state the similarity with vertices in matching sequence!

 Solved Example 1.1 • Textbook Page 49 (Example 1)

iPad Apple Pencil Workspace

In the figure: Polygon $ABCD$ has sides $AB = 12$ cm, $BC = 9$ cm, $CD = 6$ cm, $DA = 15$ cm. Polygon $EFGH$ has sides $EF = 8$ cm, $FG = 6$ cm, $GH = 4$ cm, $HE = 10$ cm. Given that corresponding angles are equal: $m(\angle A) = m(\angle E)$, $m(\angle B) = m(\angle F)$, $m(\angle C) = m(\angle G)$, $m(\angle D) = m(\angle H)$.

Prove that $ABCD \sim EFGH$ and write the similarity ratio of $ABCD$ to $EFGH$.



Polygon ABCD

Polygon EFGH

Pen Line Box Circle Axes Type Eraser

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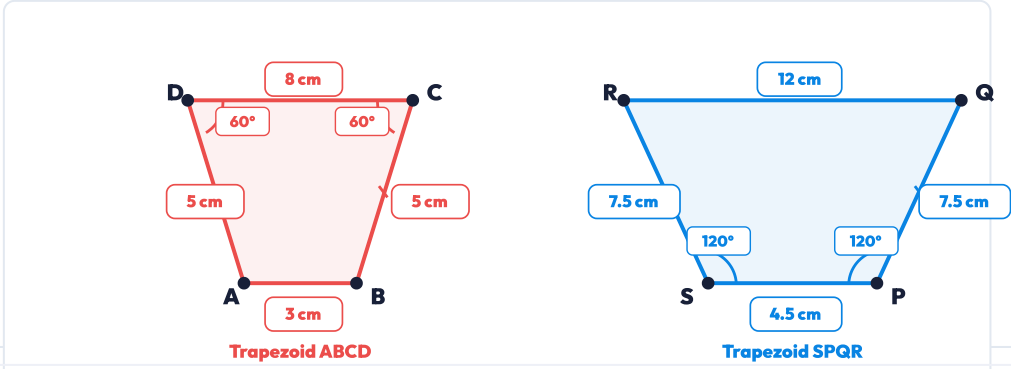
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 Solved Example 1.2 • Textbook Page 54 (Problem 13)

iPad Apple Pencil Workspace

In the figure: Isosceles Trapezoid $ABCD$ has $AB = 3$ cm, $BC = 5$ cm, $CD = 8$ cm, $DA = 5$ cm, and base angles $m(\angle C) = 60^\circ$, $m(\angle D) = 60^\circ$. Isosceles Trapezoid $SPQR$ has $SP = 4.5$ cm, $PQ = 7.5$ cm, $QR = 12$ cm, $RS = 7.5$ cm, and upper base angles $m(\angle P) = 120^\circ$, $m(\angle S) = 120^\circ$.

Prove that: Polygon $ABCD \sim$ Polygon $SPQR$.



Trapezoid ABCD

Trapezoid SPQR

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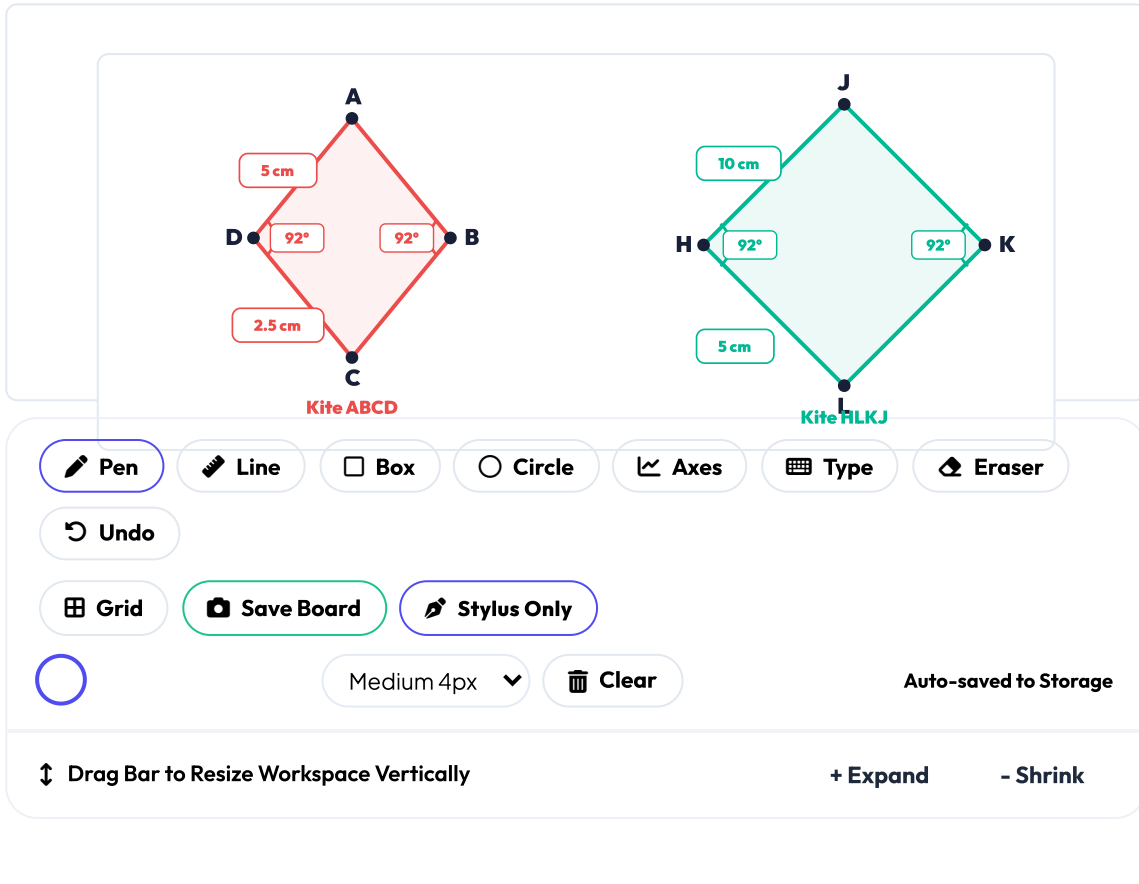
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 Self-Assessment 1 • Textbook Page 49 (Try It Yourself)

iPad Apple Pencil Workspace

In the figure: Kite $ABCD$ has $DA = 5$ cm, $AB = 5$ cm, $BC = 2.5$ cm, $CD = 2.5$ cm, with angles $m(\angle D) = 92^\circ$ and $m(\angle B) = 92^\circ$. Kite $HLKJ$ has $JH = 10$ cm, $HL = 10$ cm, $LK = 5$ cm, $KJ = 5$ cm, with angles $m(\angle L) = 92^\circ$ and $m(\angle J) = 92^\circ$. Prove that $ABCD \sim HLKJ$ and find the similarity ratio of $ABCD$ to $HLKJ$.



Concept Flashcard & Strategy Guide

Idea 2: Calculating Unknown Sides & Angles with Radical Expressions



Mastery Strategy: When polygons are stated as similar ($P_1 \sim P_2$), set up the similarity equality $\frac{\text{Side}_1}{\text{Side}_2} = \frac{\text{Side}_3}{\text{Side}_4}$. Use cross-multiplication to isolate radicals like $\sqrt{\quad}$ or $\sqrt{\quad}$. For missing angles, use the matching corresponding angle or the 360° quadrilateral sum rule!

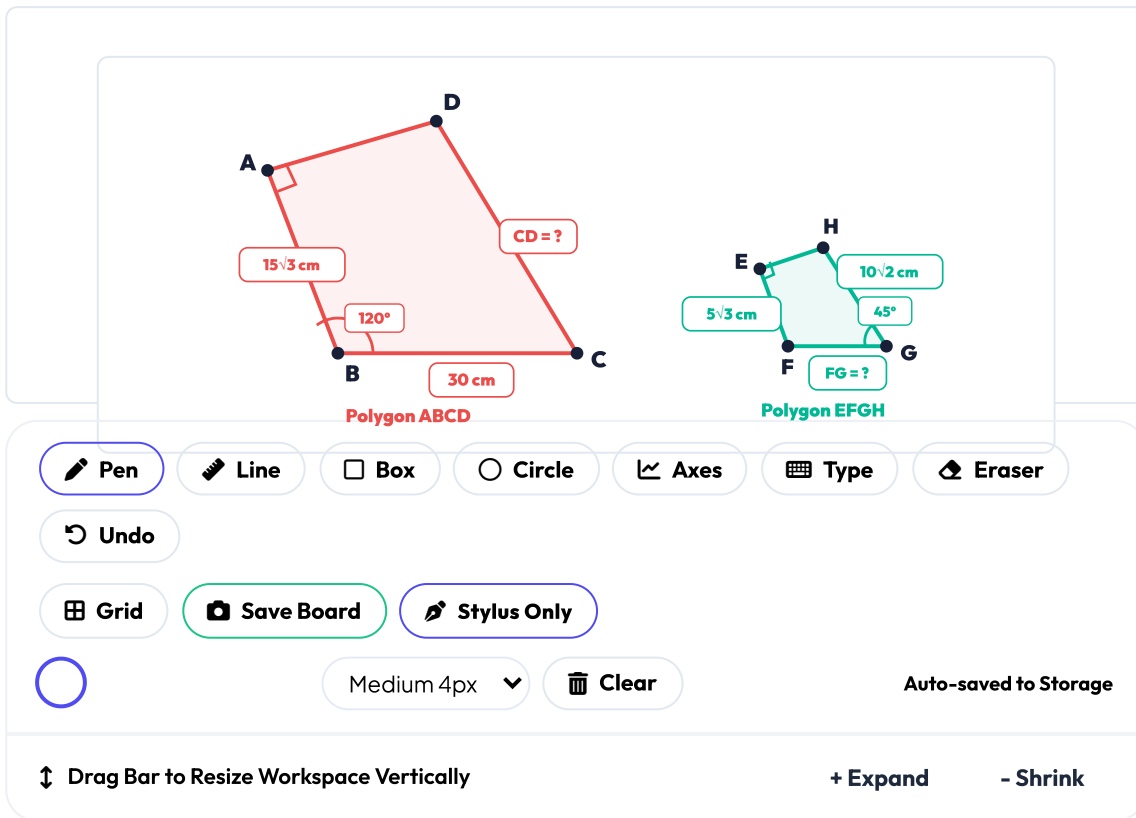
 Solved Example 2.1 • Textbook Page 49 (Example 2)

iPad Apple Pencil Workspace

Given: Polygon $ABCD \sim$ Polygon $EFGH$. In $ABCD$: $AB = 15\sqrt{3}$ cm, $BC = 30$ cm, $m(\angle B) = 120^\circ$. In $EFGH$: $EF = 5\sqrt{3}$ cm, $HG = 10\sqrt{2}$ cm, $m(\angle E) = 90^\circ$, $m(\angle F) = 120^\circ$, $m(\angle G) = 45^\circ$.

1) Find $m(\angle H)$ and $m(\angle C)$.

2) Find the lengths of FG and DC .



Polygon ABCD

Polygon EFGH

Pen Line Box Circle Axes Type Eraser

Undo

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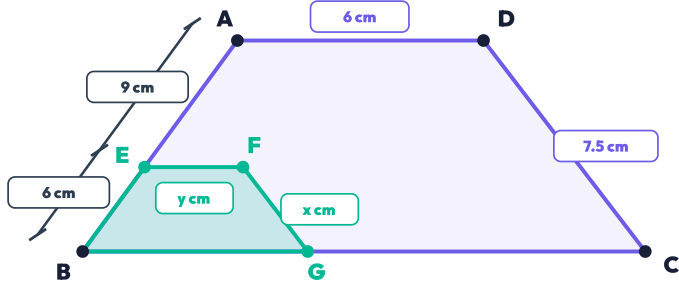
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 Solved Example 2.2 • Textbook Page 53 (Problem 10)

iPad Apple Pencil Workspace

In the figure: Polygon $ABCD \sim$ Polygon $EBGF$. $ABCD$ has $AD = 6$ cm, $CD = 7.5$ cm, and side AB has $AE = 9$ cm and $EB = 6$ cm (so $AB = 15$ cm). Polygon $EBGF$ has sides $EF = y$ cm and $FG = x$ cm.

Find the values of x and y .



Polygon $EBGF$ Polygon $ABCD$

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Drag Bar to Resize Workspace Vertically + Expand - Shrink

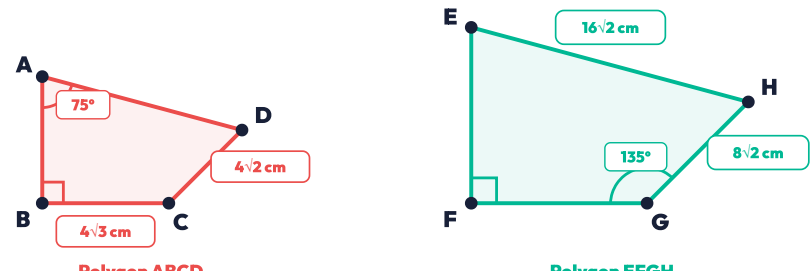
 Self-Assessment 2 • Textbook Page 49 (Try It Yourself)

iPad Apple Pencil Workspace

In the figure: Polygon $ABCD \sim$ Polygon $EFGH$. In $ABCD$: $m(\angle B) = 90^\circ$, $m(\angle A) = 75^\circ$, $BC = 4\sqrt{3}$ cm, $CD = 4\sqrt{2}$ cm. In $EFGH$: $EH = 16\sqrt{2}$ cm, $GH = 8\sqrt{2}$ cm, $m(\angle G) = 135^\circ$.

1) Find $m(\angle D)$ and $m(\angle C)$.

2) Find the lengths of AD and FG .



Polygon ABCD

Polygon EFGH

Pen Line Box Circle Axes Type Eraser

Undo

Grid Save Board Stylus Only

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Auto-saved to Storage

Drag Bar to Resize Workspace Vertically + Expand - Shrink

Concept Flashcard & Strategy Guide

Idea 3: Special Quadrilateral Similarity Criteria



Mastery Strategy: Remember the quick tests: (1) All squares are always similar. (2) Rectangles only need $\frac{L_1}{L_2} = \frac{W_1}{W_2}$. (3) Rhombuses only need one angle equal ($m(\angle 1) = m(\angle 2)$). (4) Parallelograms need one angle equal AND adjacent sides proportional! Every question is presented on its own dedicated canvas.

 Solved Example 3.1 • Rectangles Only (Textbook Page 50)

iPad Apple Pencil Workspace

Problem: Prove that Rectangle $ABCD$ (length $BC = 6$ cm, width $CD = 4$ cm) is similar to Rectangle $XYZM$ (length $YZ = 3$ cm, width $ZM = 2$ cm), and calculate the similarity ratio of $ABCD$ to $XYZM$.

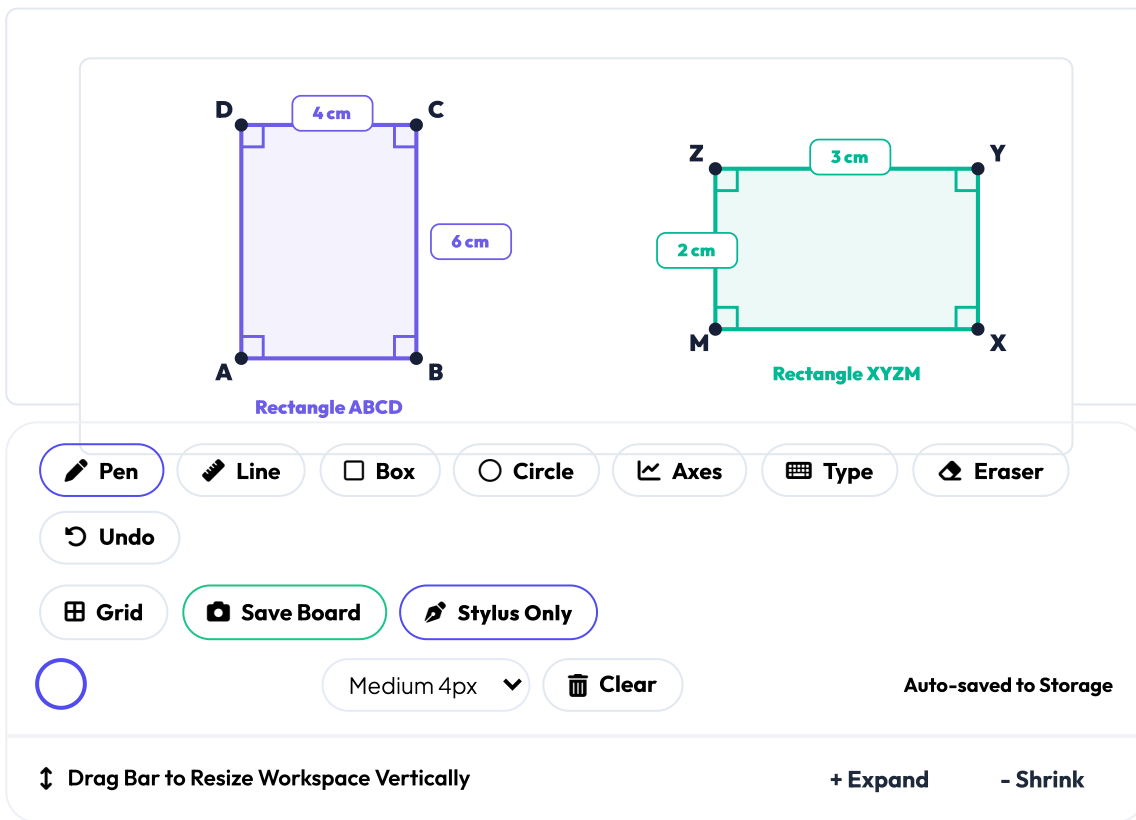


Diagram illustrating the problem setup:

- Rectangle $ABCD$** (Purple): Length $BC = 6$ cm, Width $CD = 4$ cm.
- Rectangle $XYZM$** (Green): Length $YZ = 3$ cm, Width $ZM = 2$ cm.

The workspace includes a toolbar with the following tools:

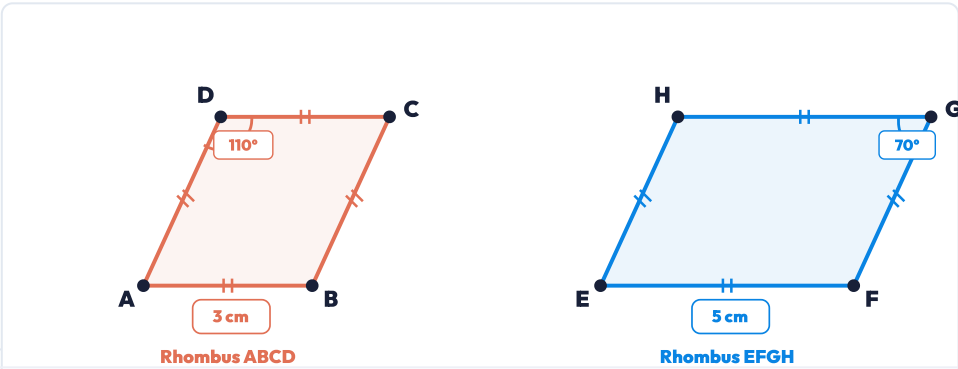
- Pen
- Line
- Box
- Circle
- Axes
- Type
- Eraser
- Undo
- Grid
- Save Board
- Stylus Only
- Medium 4px
- Clear
- Auto-saved to Storage

At the bottom, there are options to **Drag Bar to Resize Workspace Vertically**, **+ Expand**, and **- Shrink**.

 Solved Example 3.2 • Rhombuses Only (Textbook Page 50)

iPad Apple Pencil Workspace

Problem: Prove that Rhombus $ABCD$ (side length $AB = 3$ cm, angle $m(\angle D) = 110^\circ$) is similar to Rhombus $EFGH$ (side length $EF = 5$ cm, angle $m(\angle G) = 70^\circ$), and calculate the similarity ratio of $ABCD$ to $EFGH$.



Rhombus ABCD

Rhombus EFGH

Pen Line Box Circle Axes Type Eraser

Undo

Grid Save Board Stylus Only

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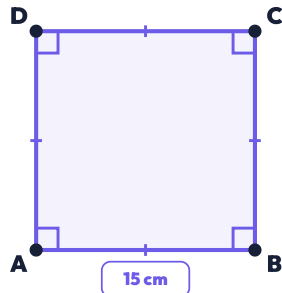
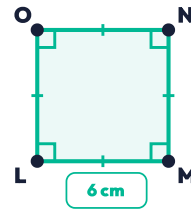
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Drag Bar to Resize Workspace Vertically + Expand - Shrink

Self-Assessment 3A • Squares Only (Textbook Page 50)

iPad Apple Pencil Workspace

Problem: Prove that Square $ABCD$ (side length 15 cm) is similar to Square $LMNO$ (side length 6 cm), and write the similarity ratio of $ABCD$ to $LMNO$.

Square $ABCD$ Square $LMNO$

Pen

Line

Box

Circle

Axes

Type

Eraser

Undo

Grid

Save Board

Stylus Only



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+ Expand

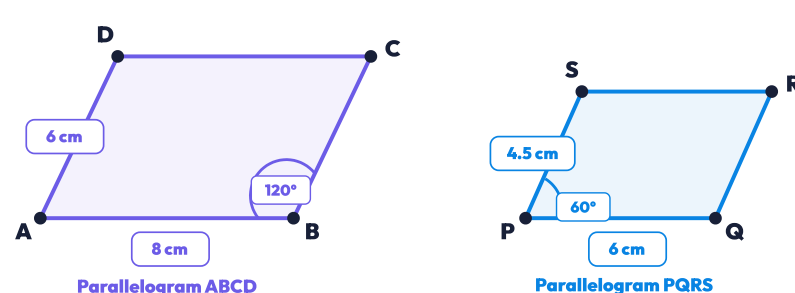
- Shrink

 Self-Assessment 3B • Parallelograms Only (Textbook Page 50)

iPad Apple Pencil Workspace

Problem: Parallelogram $ABCD$ has sides $AB = 8$ cm, $AD = 6$ cm, and angle $m(\angle B) = 120^\circ$. Parallelogram $PQRS$ has sides $PQ = 6$ cm, $PS = 4.5$ cm, and angle $m(\angle P) = 60^\circ$.

Prove that Parallelogram $ABCD \sim$ Parallelogram $PQRS$ and find the similarity ratio of $ABCD$ to $PQRS$.



Parallelogram $ABCD$ has side $AB = 8$ cm, side $AD = 6$ cm, and angle $m(\angle B) = 120^\circ$. Parallelogram $PQRS$ has side $PQ = 6$ cm, side $PS = 4.5$ cm, and angle $m(\angle P) = 60^\circ$.

Tools: Pen, Line, Box, Circle, Axes, Type, Eraser, Undo, Grid, Save Board, Stylus Only, Medium 4px, Clear, Auto-saved to Storage.

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Concept Flashcard & Strategy Guide

Idea 4: Ratio of Perimeters & Solving Algebraic Systems
(x, y, z)



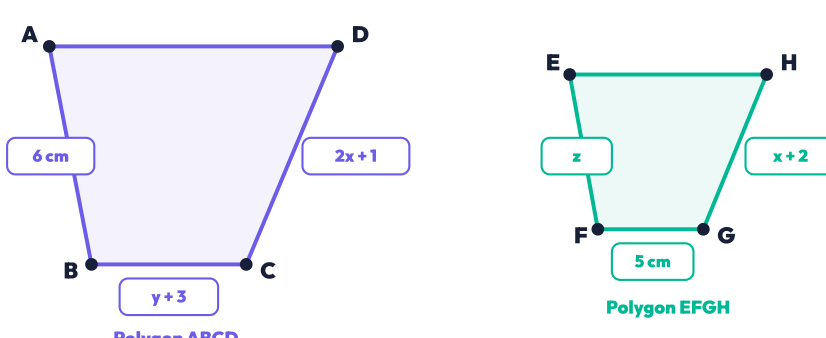
Perimeter Theorem: The ratio of the perimeters of two similar polygons is equal to the similarity ratio k of any two corresponding sides: $\frac{\text{Perimeter}_1}{\text{Perimeter}_2} = k$.
Equate each side expression to k to solve for algebraic variables x, y, z !

 Solved Example 4.1 • Textbook Page 51 (Example 4)

iPad Apple Pencil Workspace

Given: Polygon $ABCD \sim$ Polygon $EFGH$, and $\frac{\text{Perimeter of } ABCD}{\text{Perimeter of } EFGH} = \frac{3}{2}$. Sides given: $AB = 6$ cm, $EF = z$ cm; $BC = (y + 3)$ cm, $FG = 5$ cm; $CD = (2x + 1)$ cm, $GH = (x + 2)$ cm.

Find the values of x, y, z .



Polygon ABCD

Polygon EFGH

Pen Line Box Circle Axes Type Eraser

Undo

Grid Save Board Stylus Only

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Auto-saved to Storage

Drag Bar to Resize Workspace Vertically + Expand - Shrink

 Solved Example 4.2 • Textbook Page 53 (Problem 12)

iPad Apple Pencil Workspace

Two polygons are similar. The side lengths of the first polygon are 3 cm, 5 cm, 6 cm, 8 cm, 10 cm. The perimeter of the second polygon is 80 cm.

Find the lengths of the five sides of the second polygon.

Pen Line Box Circle Axes Type Eraser

Undo

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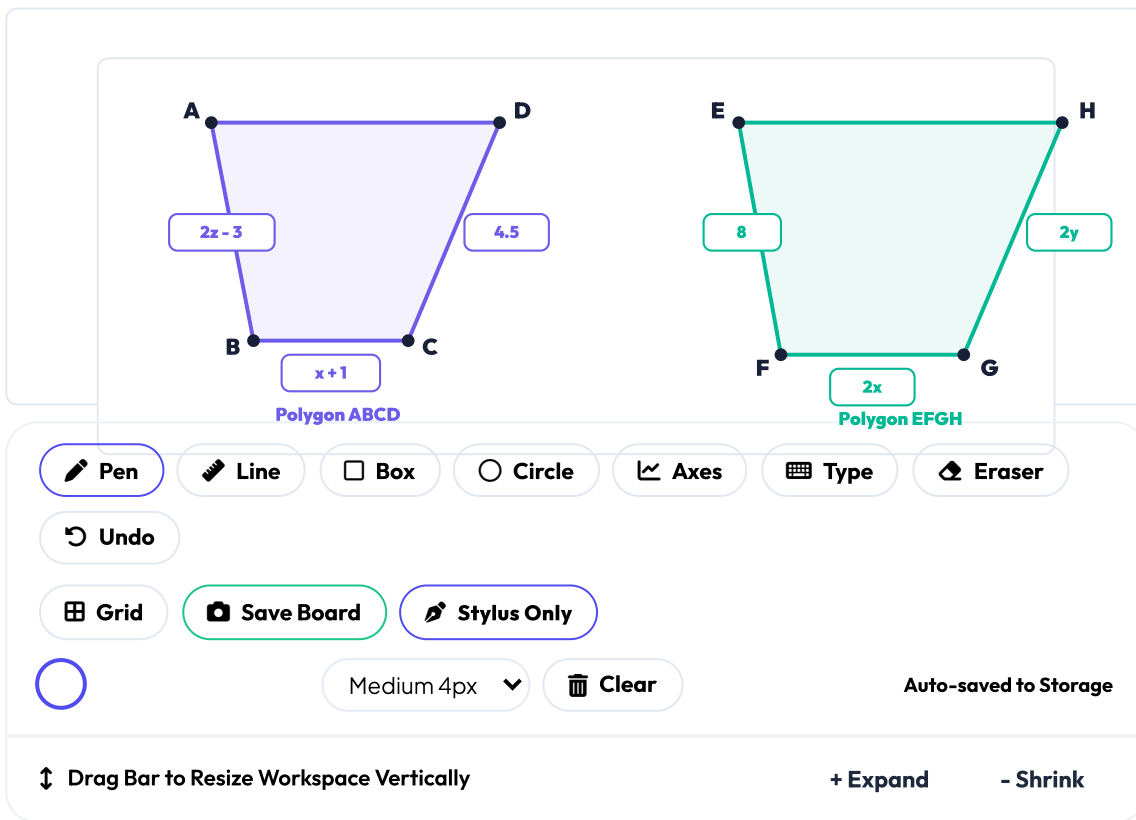
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 Self-Assessment 4 • Textbook Page 51 (Try It Yourself)

iPad Apple Pencil Workspace

Given: Polygon $ABCD \sim$ Polygon $EFGH$, and $\frac{\text{Perimeter of } ABCD}{\text{Perimeter of } EFGH} = \frac{3}{4}$. In $ABCD$: $AB = 2z - 3$, $BC = x + 1$, $CD = 4.5$. In $EFGH$: $EF = 8$, $FG = 2x$, $GH = 2y$. Find the values of x, y, z .



Polygon ABCD

Polygon EFGH

Pen Line Box Circle Axes Type Eraser

Undo

Grid Save Board Stylus Only

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Auto-saved to Storage

Drag Bar to Resize Workspace Vertically + Expand - Shrink

 Self-Assessment Reflection

How confident do you feel with Lesson 3-1: Similarity of Polygons? Click your current level:



Need Practice



Getting It



Good Understanding



Mastered Confidently

Mr Ahmed Abd El-Motaal

Math teacher and content creator

Empowering middle and high school students with engaging, interactive, visual mathematics. Dedicated to precision proofs, step-by-step clarity, and top exam preparation.

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- > 3. Timed Quiz - 10 Marks (3 Models)

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